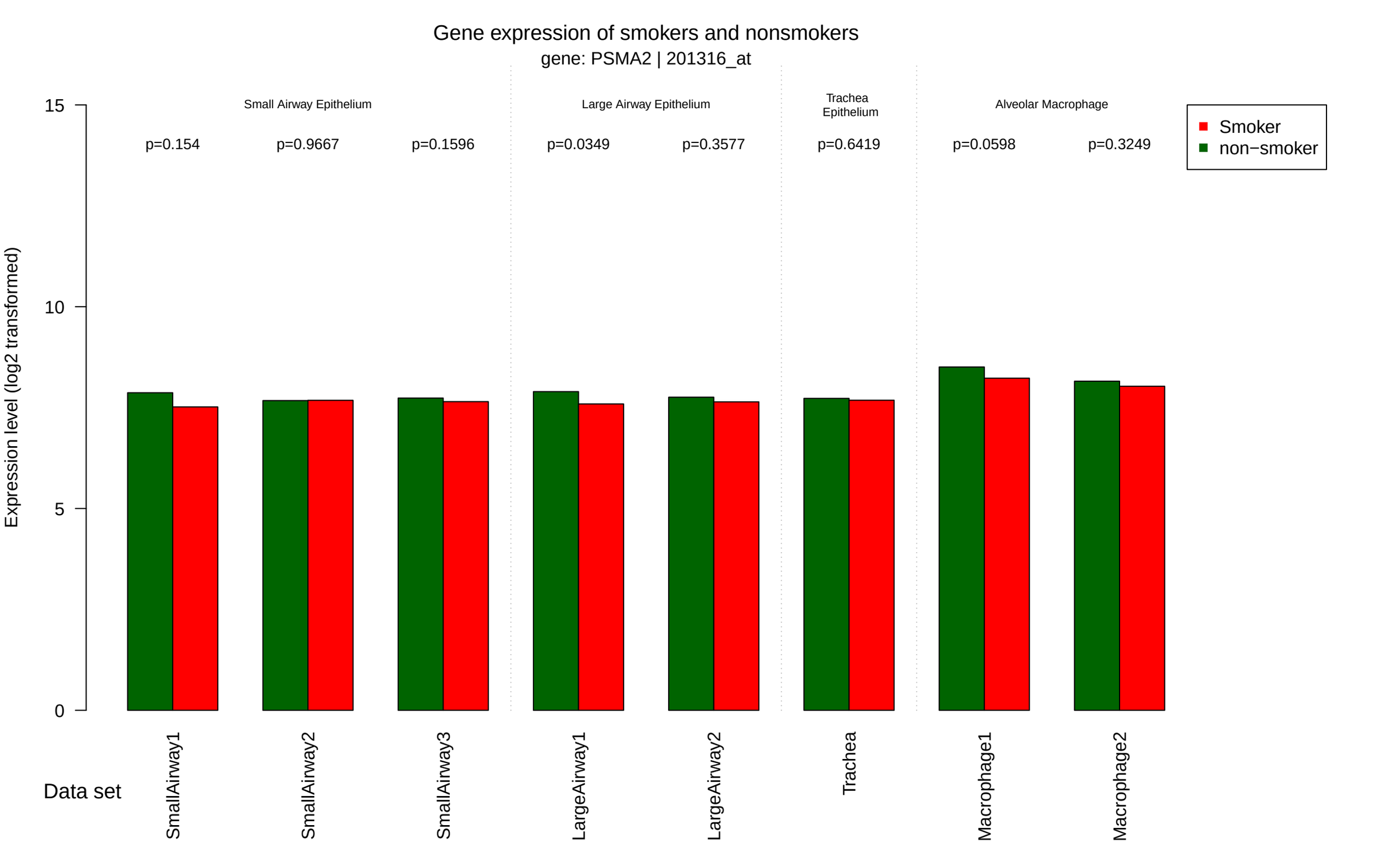
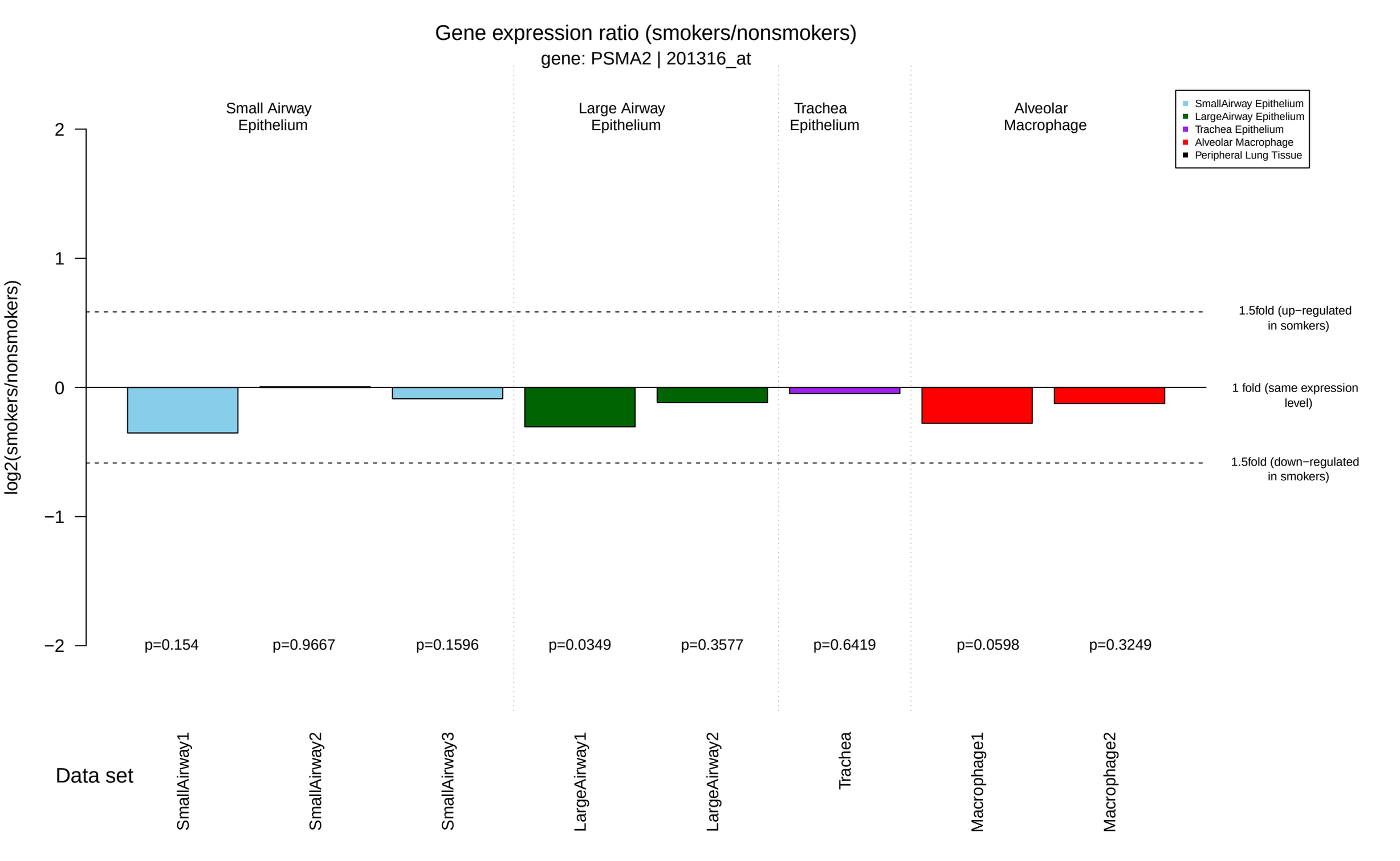
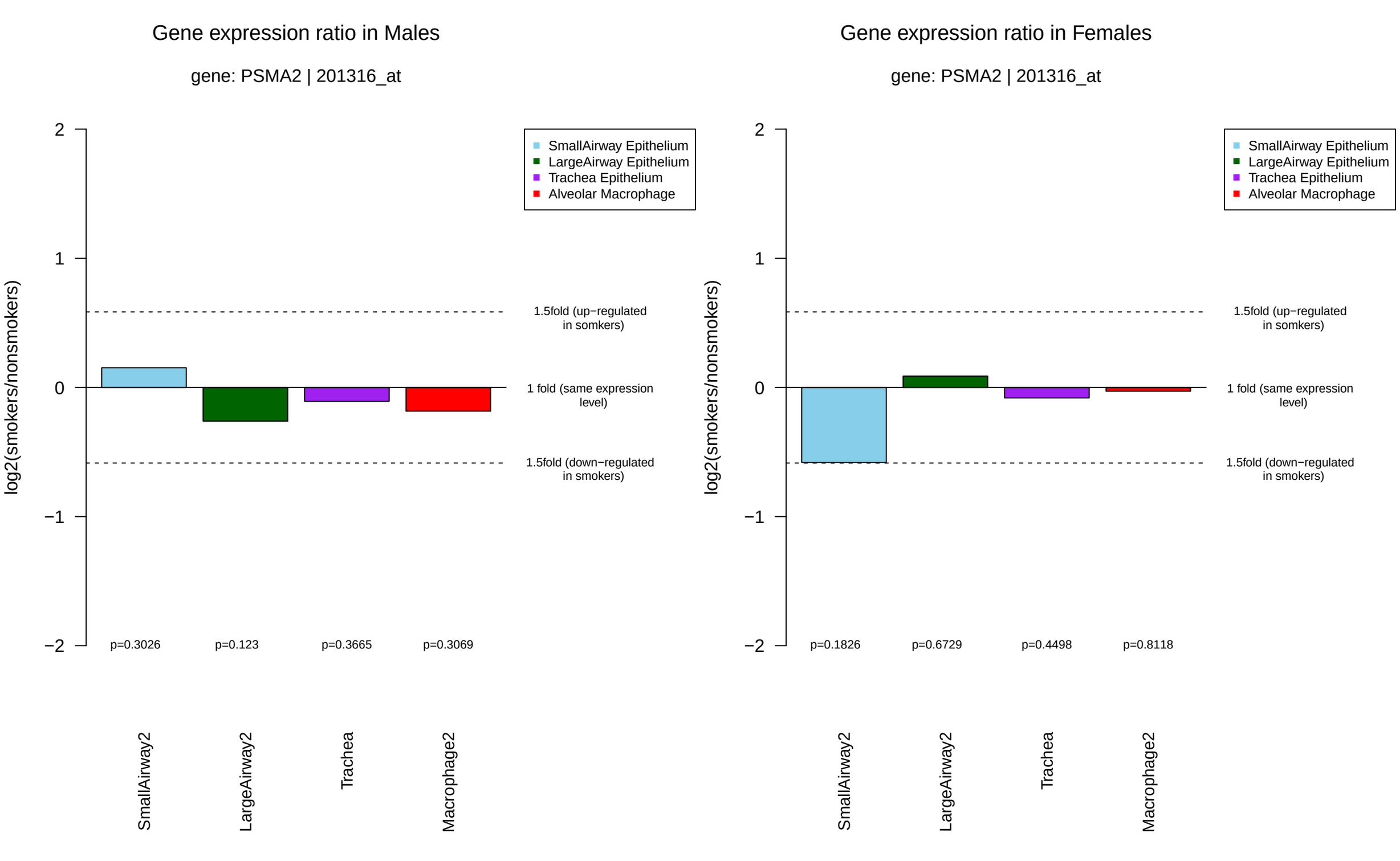
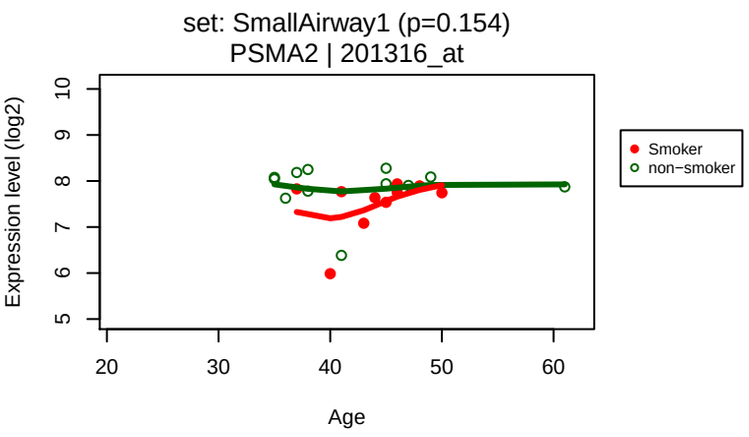
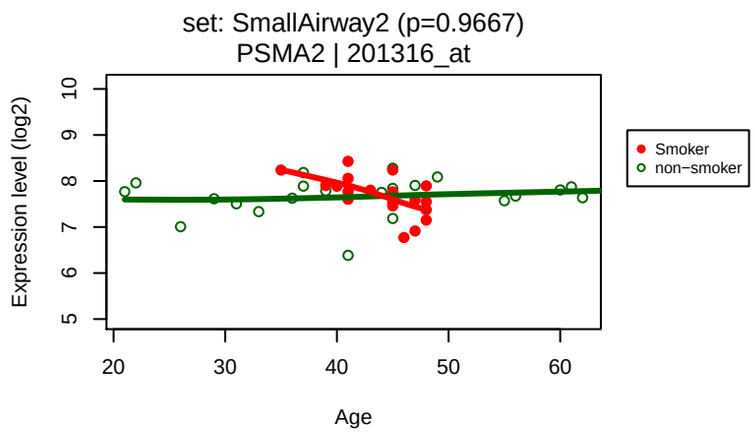
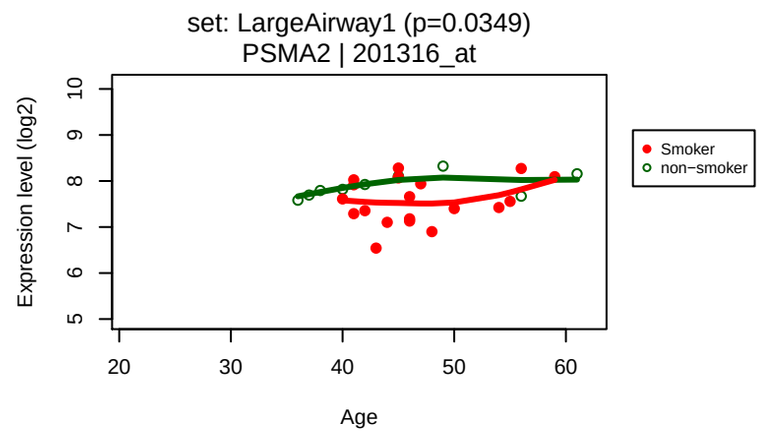

Gene expression of smokers and nonsmokers
gene: PSMA2 | 201316_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.154
p=0.9667
p=0.1596
p=0.0349
p=0.3577
p=0.6419
p=0.0598
p=0.3249
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

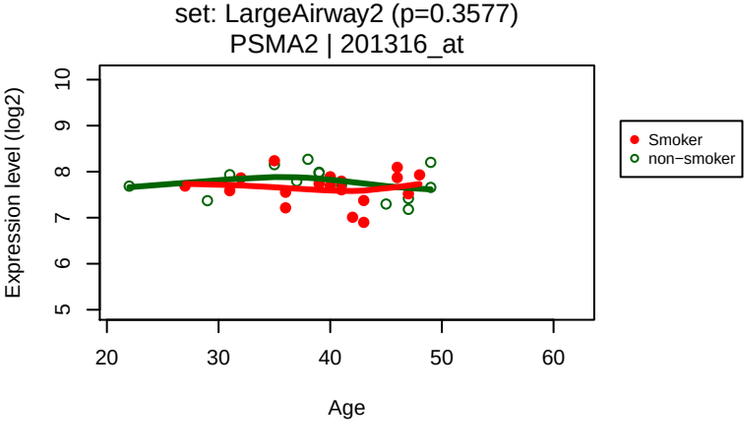

Gene expression ratio (smokers/nonsmokers)
gene: PSMA2 | 201316_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.154
p=0.9667
p=0.1596
p=0.0349
p=0.3577
p=0.6419
p=0.0598
p=0.3249
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

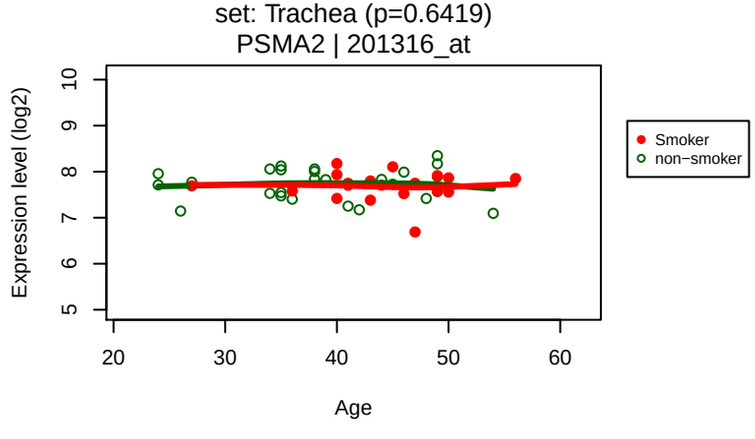

Gene expression ratio in Males
gene: PSMA2 | 201316_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3026
p=0.123
p=0.3665
p=0.3069
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSMA2 | 201316_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1826
p=0.6729
p=0.4498
p=0.8118
SmallAirway2
LargeAirway2
Trachea
Macrophage2

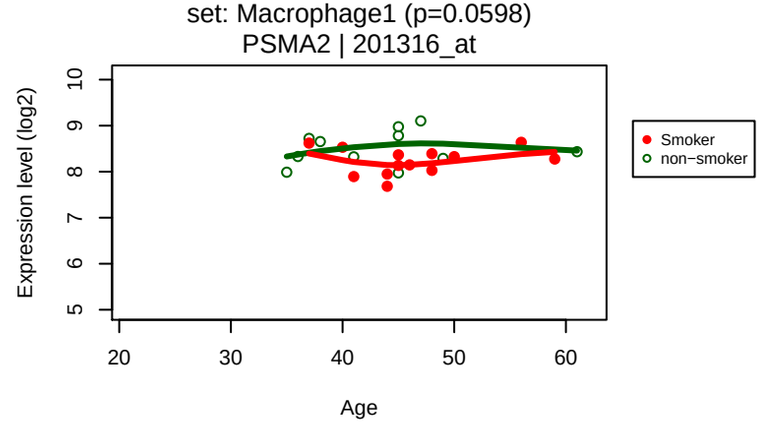

set: SmallAirway1 (p=0.154)
PSMA2 | 201316_at
Expression level (log2)
Age
Smoker
non−smoker

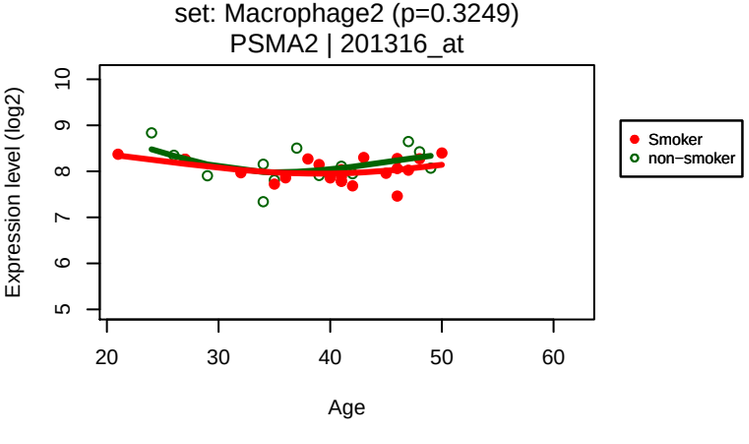

set: SmallAirway2 (p=0.9667)
PSMA2 | 201316_at
Expression level (log2)
Age
Smoker
non−smoker

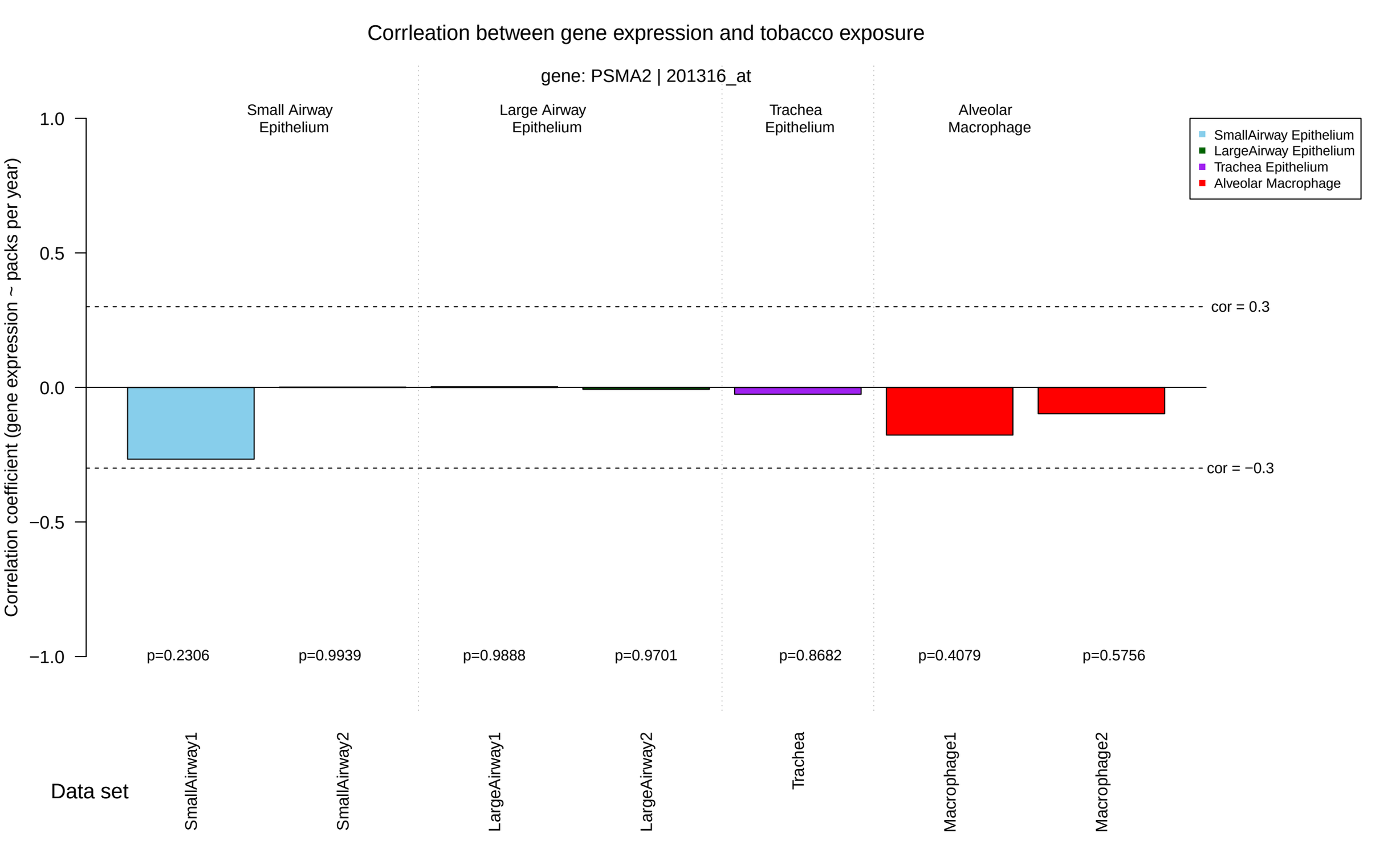

set: LargeAirway1 (p=0.0349)
PSMA2 | 201316_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3577)
PSMA2 | 201316_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6419)
PSMA2 | 201316_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0598)
PSMA2 | 201316_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3249)
PSMA2 | 201316_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSMA2 | 201316_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2306
p=0.9939
p=0.9888
p=0.9701
p=0.8682
p=0.4079
p=0.5756
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2